

RevPlay – Application Architecture Diagram

RevPlay follows a modular, layered architecture that separates responsibilities across different layers, ensuring maintainability and scalability.

Architecture Layers

Layer	Responsibilities	Components
Presentation Layer (UI)	Handles console input/output and menu navigation.	Main.java
Service Layer	Implements business logic, ownership checks, and orchestration.	AuthService, SongService, AlbumService, PlaylistService
DAO Layer	Handles database operations using JDBC and SQL.	UserDao, SongDao, AlbumDao, PlaylistDao
Database Layer	Stores persistent data with normalized schema.	MySQL Database

Architecture Flow

Console UI → Service Layer → DAO Layer → MySQL Database

This layered design ensures loose coupling, easier debugging, and a clear upgrade path to REST-based or microservices architectures.